

Table 1: Amino acid substitutions for FNV strains, relative to the Parental French viscerotropic virus.

| Nucleotide Position | Gene | Codon | FVV | Asibi | FNV-Dakar | FNV-IP | FNV-Yale | FNV-FC | FNV-NT         |
|---------------------|------|-------|-----|-------|-----------|--------|----------|--------|----------------|
| 275                 | C    | 53    | F   | -     | -         | -      | -        | -      | L              |
| 310                 | C    | 64    | H   | -     | Q         | -      | -        | -      | -              |
| 357 <sup>a</sup>    | C    | 80    | V   | -     | A         | A      | A        | A      | A              |
| 386 <sup>a</sup>    | C    | 90    | S   | -     | G         | G      | G        | G      | G              |
| 491                 | prM  | 4     | V   | -     | M         | -      | -        | -      | -              |
| 554 <sup>a</sup>    | prM  | 25    | V   | -     | M         | M      | M        | M      | M              |
| 596                 | prM  | 39    | Y   | -     | -         | -      | -        | -      | H              |
| 701                 | prM  | 74    | A   | -     | T         | -      | -        | -      | -              |
| 854 <sup>ab</sup>   | M    | 36    | L   | -     | F         | F      | F        | F      | F              |
| 989                 | E    | 6     | I   | -     | V         | -      | -        | -      | -              |
| 992 <sup>a</sup>    | E    | 7     | T   | -     | A         | A      | A        | A      | - <sup>c</sup> |
| 1134 <sup>a</sup>   | E    | 54    | A   | -     | V         | V      | V        | V      | V              |
| 1184                | E    | 71    | N   | -     | D         | -      | -        | -      | -              |
| 1392                | E    | 140   | R   | -     | K         | -      | -        | -      | K              |
| 1395                | E    | 141   | A   | -     | -         | -      | -        | -      | V              |
| 1424                | E    | 151   | N   | -     | D         | -      | -        | -      | D              |
| 1432 <sup>a</sup>   | E    | 153   | N   | -     | K         | K      | K        | K      | - <sup>c</sup> |
| 1437                | E    | 155   | D   | -     | -         | -      | -        | -      | A              |
| 1440                | E    | 156   | I   | -     | -         | -      | -        | -      | T              |
| 1461                | E    | 163   | A   | -     | V         | -      | -        | -      | -              |
| 1506                | E    | 178   | A   | -     | V         | -      | -        | -      | -              |
| 1572 <sup>ab</sup>  | E    | 200   | T   | K     | K         | K      | K        | K      | K              |
| 1718 <sup>a</sup>   | E    | 249   | N   | -     | D         | D      | D        | D      | D              |
| 1965 <sup>ab</sup>  | E    | 331   | K   | -     | R         | R      | R        | R      | R              |
| 2105                | E    | 378   | V   | -     | I         | -      | -        | -      | -              |
| 2193                | E    | 407   | A   | -     | -         | -      | -        | -      | V              |
| 2344 <sup>a</sup>   | E    | 457   | M   | -     | -         | I      | I        | I      | - <sup>c</sup> |
| 3234                | NS1  | 261   | K   | -     | -         | -      | -        | -      | R              |
| 3465                | NS1  | 338   | R   | -     | M         | -      | -        | -      | -              |
| 3469                | NS1  | 339   | K   | -     | -         | -      | -        | -      | N              |
| 3596                | NS2A | 30    | M   | -     | V         | -      | -        | -      | -              |
| 3752 <sup>a</sup>   | NS2A | 82    | I   | -     | V         | V      | V        | V      | V              |
| 3821 <sup>a</sup>   | NS2A | 105   | T   | -     | A         | A      | A        | A      | A              |
| 3947                | NS2A | 147   | T   | -     | A         | -      | -        | -      | -              |
| 4212                | NS2B | 11    | A   | -     | -         | -      | -        | -      | V              |
| 4358 <sup>a</sup>   | NS2B | 60    | S   | -     | A         | A      | A        | A      | A              |
| 4464                | NS2B | 95    | D   | -     | -         | -      | -        | -      | G              |
| 4550                | NS2B | 124   | H   | -     | -         | -      | -        | -      | Y              |
| 4559                | NS2B | 127   | G   | -     | R         | -      | -        | -      | -              |
| 4607 <sup>a</sup>   | NS3  | 13    | I   | -     | -         | V      | V        | V      | - <sup>c</sup> |
| 4754                | NS3  | 62    | R   | -     | W         | -      | -        | -      | W              |
| 4878                | NS3  | 103   | V   | -     | A         | -      | -        | -      | A              |
| 5145 <sup>a</sup>   | NS3  | 192   | M   | -     | K         | K      | K        | K      | K              |
| 5361                | NS3  | 264   | A   | -     | V         | -      | -        | -      | -              |
| 5408                | NS3  | 280   | I   | V     | -         | -      | -        | -      | -              |
| 6123                | NS3  | 518   | T   | -     | I         | -      | -        | -      | -              |
| 6451                | NS4A | 4     | E   | -     | D         | -      | -        | -      | -              |
| 6459                | NS4A | 7     | V   | -     | -         | -      | -        | -      | A              |
| 6527 <sup>a</sup>   | NS4A | 29    | F   | -     | -         | L      | L        | L      | L              |
| 6602                | NS4A | 54    | M   | -     | -         | -      | -        | -      | L              |
| 6611                | NS4A | 57    | I   | -     | L         | -      | -        | -      | -              |
| 6761                | NS4A | 107   | M   | -     | -         | -      | -        | -      | L              |

|                    |       |     |   |   |   |   |   |   |                |
|--------------------|-------|-----|---|---|---|---|---|---|----------------|
| 6843 <sup>a</sup>  | NS4A  | 135 | Y | - | F | F | F | F | F              |
| 6875               | NS4A  | 145 | V | - | - | - | - | - | L              |
| 6942 <sup>a</sup>  | NS4B  | 19  | L | - | - | S | S | S | - <sup>c</sup> |
| 6953               | NS4B  | 23  | S | - | G | - | - | - | G              |
| 7171 <sup>ab</sup> | NS4B  | 95  | I | - | M | M | M | M | M              |
| 7178               | NS4B  | 98  | V | - | - | I | I | I | I              |
| 7380               | NS4B  | 165 | A | - | - | V | V | V | -              |
| 7601               | NS4B  | 239 | M | - | V | - | - | - | -              |
| 7602               | NS4B  | 239 | M | - | T | - | - | - | -              |
| 7629               | NS4B  | 248 | G | - | A | - | - | - | -              |
| 7641 <sup>a</sup>  | NS5   | 2   | R | - | T | T | T | T | - <sup>c</sup> |
| 7642               | NS5   | 2   | R | - | - | - | - | - | S              |
| 7694               | NS5   | 20  | D | - | N | - | - | - | N              |
| 8030               | NS5   | 132 | I | - | V | - | - | - | V              |
| 8090               | NS5   | 152 | S | - | - | - | - | - | P              |
| 8409 <sup>a</sup>  | NS5   | 258 | I | - | T | T | T | T | T              |
| 8471               | NS5   | 279 | E | - | K | - | - | - | -              |
| 8618               | NS5   | 328 | I | - | L | - | - | - | -              |
| 8640 <sup>a</sup>  | NS5   | 335 | R | - | K | K | K | K | K              |
| 8890               | NS5   | 418 | E | - | D | - | - | - | -              |
| 8928               | NS5   | 431 | K | - | - | - | - | - | R              |
| 8939               | NS5   | 435 | L | - | - | - | - | - | M              |
| 9227 <sup>a</sup>  | NS5   | 531 | F | - | L | L | L | L | L              |
| 9605               | NS5   | 657 | N | - | D | - | - | - | D              |
| 9615               | NS5   | 660 | K | - | R | - | R | - | R              |
| 10074              | NS5   | 813 | M | - | R | - | - | - | -              |
| 10120              | NS5   | 828 | M | - | - | - | - | - | I              |
| 10232              | NS5   | 866 | H | - | N | - | - | - | -              |
| 10268              | NS5   | 878 | I | - | V | - | V | - | V              |
| 10338              | NS5   | 901 | P | - | - | L | - | L | -              |
| 10357              | 3'UTR | -   | C | - | U | - | U | - | U              |
| 10358              | 3'UTR | -   | A | - | G | - | G | - | G              |
| 10363              | 3'UTR | -   | C | - | - | - | - | - | U              |
| 10367              | 3'UTR | -   | U | - | - | C | - | C | -              |
| 10389              | 3'UTR | -   | C | - | U | - | - | - | -              |
| 10394              | 3'UTR | -   | G | - | A | - | - | - | -              |
| 10404              | 3'UTR | -   | G | - | - | - | A | - | -              |
| 10411              | 3'UTR | -   | C | - | A | - | - | - | -              |
| 10418              | 3'UTR | -   | U | - | - | C | - | C | -              |
| 10528              | 3'UTR | -   | G | - | C | - | - | - | -              |
| 10531              | 3'UTR | -   | G | - | A | - | - | - | -              |
| 10550              | 3'UTR | -   | U | - | - | C | - | C | -              |
| 10554              | 3'UTR | -   | C | - | - | - | - | - | U              |
| 10798              | 3'UTR | -   | A | - | G | - | G | - | G              |
| 10800              | 3'UTR | -   | A | G | - | - | - | - | -              |

<sup>a</sup> Conserved at least 75 percent for FNV-IP, FNV-Yale, FNV-FC, and FNV-NT strains.

<sup>b</sup> Intersection of sequence with the Asibi strain or is observed in 17D attenuation.

<sup>c</sup> Not conserved in FNV-NT.